CLINICAL · NOSE & SINUSES

Facial plastic surgery

Facial plastic surgery begins with analysis: proportions, angles, the skin and the structures beneath it. It then asks what the patient wants and whether that is achievable and wise. For the ENT surgeon the core is the nose, where form and airway must be fixed together. The other procedures are the ear, the ageing face and the injectables, each with its own anatomy and its own complications to avoid.

AT A GLANCE

- Major tip supports: lower lateral cartilages, footplate to caudal septum, scroll.
- Spreader grafts for the internal valve; batten grafts for the external valve.
- Retrobulbar haematoma after blepharoplasty: lateral canthotomy now.
- Mustarde sutures make the antihelical fold; Furnas sets back the concha.
- Preservation rhinoplasty lowers the dorsum without resecting it; screen for body dysmorphic disorder.

1 CHAPTER ONE Facial analysis

Measure before you cut.

1.1 Landmarks

Analysis uses a small vocabulary of points. In the midline from above down lie the **trichion** (the hairline), the **glabella** (the most prominent point of the forehead) and the **nasion**, the deepest point of the nasofrontal angle at the nasofrontal suture. On the dorsum, the **rhinion** marks the junction of bone and cartilage, where the skin is thinnest, and the light reflexes of the domes are the **tip-defining points**. The **subnasale** is where the columella meets the lip and the **stomion** where the lips meet. On the chin, the **pogonion** is the most anterior point and the **menton** the lowest, and the **cervical point** is the deepest point between the submental region and the neck.

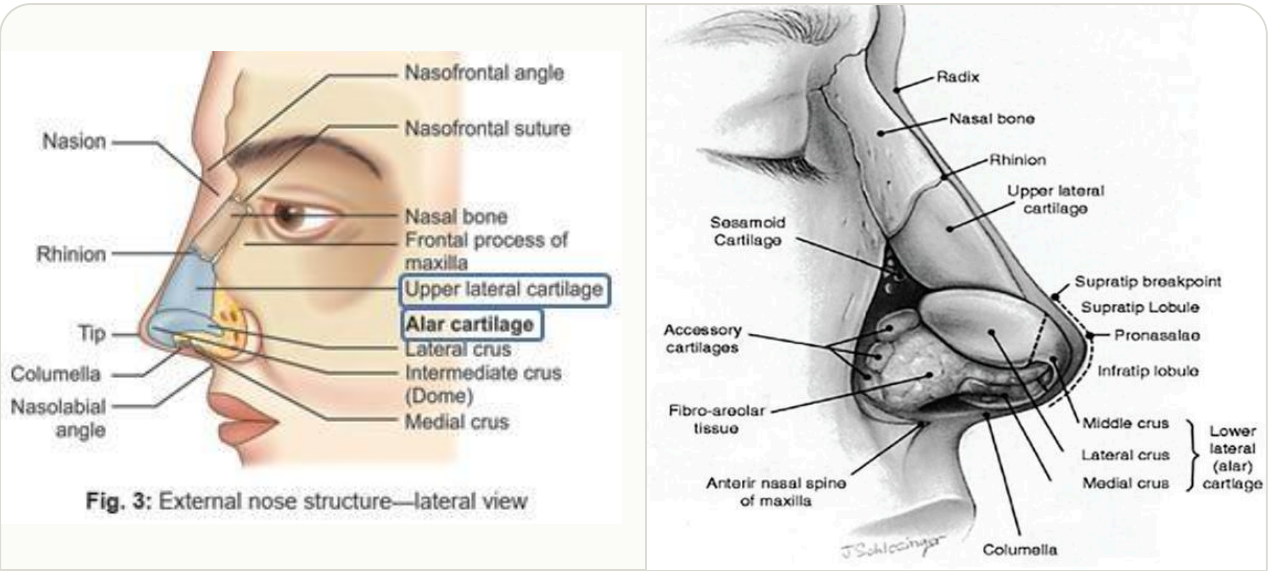


FIGURE 1 Surface landmarks and the osteocartilaginous framework of the external nose.
Anatomy atlas · ENT Steps p. 92, ENT Steps p. 93

1.2 Proportions

In height the face divides into **thirds**: trichion to glabella, glabella to subnasale, and subnasale to menton are about equal, and within the lower third the upper lip takes a third and the lower lip and chin two-thirds. In width it divides into **fifths**, each about one eye's width; the intercanthal distance equals an eye width and also the width of the nasal base. Width to height is about 3:4.

FACIAL ANGLES

ANGLE	DEFINED BY	IDEAL
Nasofrontal	Glabella–nasion and nasion–tip	115 to 130°
Nasofacial	Facial plane (glabella–pogonion) and the dorsum	30 to 40°
Nasolabial	Columella and upper lip at subnasale	90 to 95° in men, 95 to 110° (up to 120°) in women
Nasomental	Dorsum and tip–pogonion	120 to 132°
Mentocervical	Facial plane and the submental line	80 to 95°

1.3 The chin and occlusion

In profile the chin should sit on a vertical line dropped from the lower lip, or just behind it in women. A weak chin is either **microgenia**, a small chin with normal occlusion that is corrected with an implant or a *sliding genioplasty*, or **retrognathia**, a retruded mandible with a class II bite that needs orthognathic surgery such as a sagittal split osteotomy. The distinction is made by looking at the bite.

Angle's classification describes occlusion by the upper first molar. In **class I**, the normal state, its mesiobuccal cusp sits in the buccal groove of the lower first molar; in **class II** the upper molar sits forward of the lower, as in retrognathia; and in **class III** it sits behind, as in prognathism.

Chin implants of silastic, porous polyethylene or ePTFE are placed in a subperiosteal pocket through a submental or intraoral incision, and can displace, become infected, cause bone resorption or injure the mental nerve. A sliding genioplasty adds both projection and height, at the risk of injury to the mental nerve and the tooth roots.

1.4 Skin

The **Fitzpatrick types** run from I, which always burns and never tans, to VI, which never burns; types III to VI are at more risk of pigment change after peels and lasers. **Glogau's photoageing classes** progress from I (no wrinkles) through II (wrinkles in motion) and III (wrinkles at rest) to IV (only wrinkles). Nasal skin is thick and sebaceous at the tip and supratip and thinnest at the rhinion: thick skin hides tip work and swells for months, while thin skin shows every irregularity.

CHAPTER TWO

2 Rhinoplasty

Form and function together.

2.1 Surgical anatomy

The nose is best thought of in thirds. The **upper, bony third** is the nasal bones. The **middle third** is the upper lateral cartilages fused to the dorsal septum, which together form the *internal nasal valve*, normally an angle of 10 to 15°. The **lower third** is the lower lateral (alar) cartilages, with their medial, intermediate and lateral crura and the domes, and the external valve. The junction of the upper and lower laterals is the **scroll**, and the area where the nasal bones meet the upper lateral cartilages and septum is the **keystone**, whose disruption lets the middle vault collapse.

Tip support comes from three major mechanisms: the size, shape and strength of the lower lateral cartilages; the attachment of the medial crural footplates to the caudal septum; and the attachment of the upper to the lower laterals at the scroll. The minor mechanisms are the interdomal ligament, the dorsal cartilaginous septum, the sesamoid complex, the skin and soft-tissue envelope, the nasal spine and the membranous septum. Every incision or resection that disrupts one of these must be compensated for.

The **tripod concept** makes tip surgery predictable. The two lateral crura and the conjoined medial crura form the three legs of a tripod; shortening or lengthening the legs changes *projection* and *rotation* in ways that can be planned.

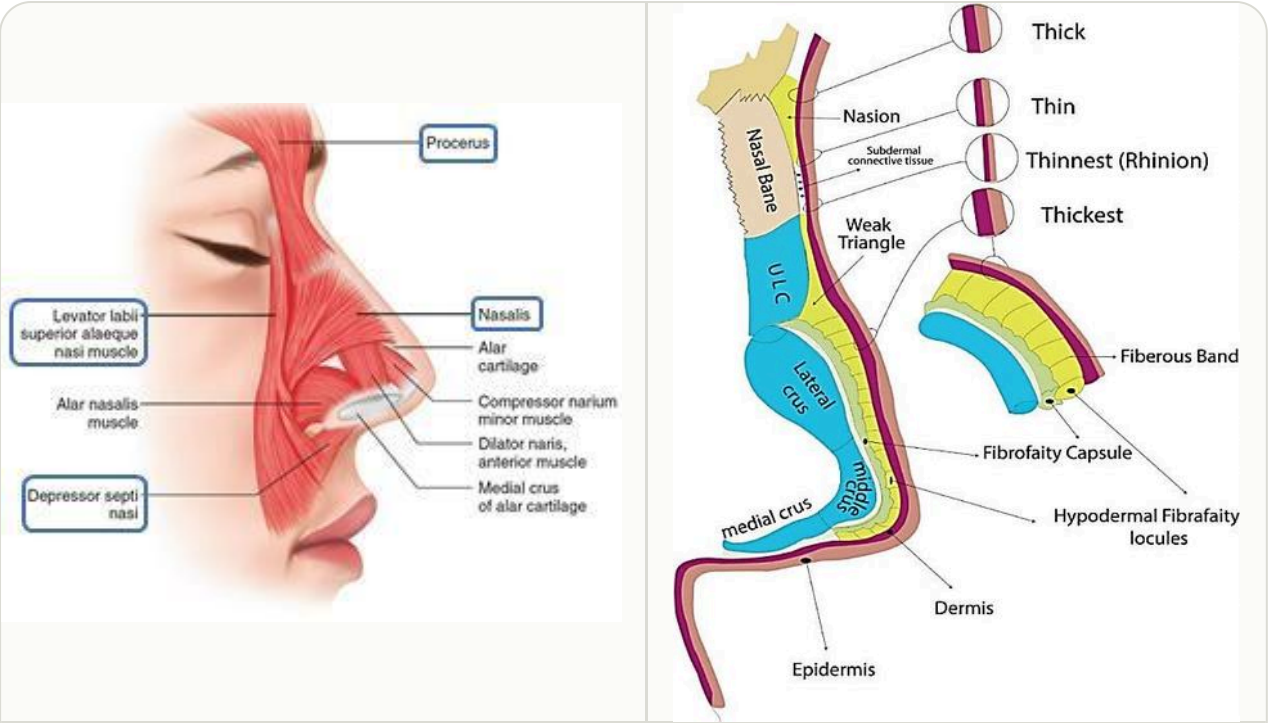


FIGURE 2 Muscles of the nose, and nasal skin thickness from nasion to tip.
Anatomy atlas · ENT Steps p. 95, ENT Steps p. 96

2.2 Analysis of the nose

On the **frontal view**, assess deviation by thirds, the width of the bony base compared with the alar base, smooth *brow–tip aesthetic lines* running from the medial brow to the tip-defining point, tip definition, and the "gull in flight" outline of the alae and columella.

On **profile**, note the radix position, which should lie between the lash line and the upper lid crease, any dorsal hump, and a supratip break, which is desirable in women. Tip *projection* is judged by the *Goode ratio*, in which projection is 0.55 to 0.6 of nasal length, or by the 3:4:5 triangle; *rotation* is read from the nasolabial angle; and 2 to 4 mm of *columellar show* is normal.

On the **base view**, the nose should form an equilateral triangle with a columella-to-lobule ratio of 2:1, symmetrical nostrils and no caudal septal deviation. **Inside the nose**, examine the septum, turbinates and valves; the *Cottle manoeuvre*, in which lateral traction on the cheek improves the airway, suggests valve collapse, and the modified Cottle test with a cotton bud localises it.

Take standard photographs, and always weigh the patient as well as the nose: body dysmorphic disorder and unrealistic expectations are the commonest causes of an unhappy outcome after technically sound surgery.

2.3 Approaches and incisions

The incisions are the **marginal**, along the caudal edge of the lateral crus; the **intercartilaginous**, between the upper and lower laterals; the **transcartilaginous**, through the lateral crus; the **transcolumellar**, an inverted V or stair step; and the **transfixion or hemitransfixion**, between the caudal septum and the medial crura.

Endonasal (closed) rhinoplasty uses either a non-delivery approach, cartilage-splitting or retrograde, or a delivery approach in which the alar cartilages are brought out as bipedicled flaps. It leaves no external scar and causes less swelling, but exposure is limited.

External (open) rhinoplasty joins transcolumellar and marginal incisions and lifts the whole skin envelope. It gives the best exposure for precise sutures and grafts and is preferred for revision, cleft and crooked noses, major tip work and teaching. Its costs are a columellar scar, more oedema and the disruption of a minor tip support.

COMMON MANOEUVRES

PROBLEM	MANOEUVRE
Dorsal hump	Component reduction: rasp the bone, cut the cartilage; then osteotomies to close the open roof; spreader grafts to keep the middle vault and valve
Wide or crooked bony vault	Medial, lateral (low-to-high along the nasomaxillary groove, starting above the inferior turbinate attachment) and intermediate osteotomies
Crooked middle third	Septoplasty, asymmetric spreader grafts or spreader flaps, camouflage onlay grafts
Bulbous tip	Cephalic trim, keeping at least 6 to 10 mm of lateral crus; dome-binding (transdomal, interdomal) sutures; lateral crural strut grafts
Under-projected tip	Columellar strut, caudal septal extension graft, dome sutures, tip (shield, cap) grafts, lateral crural steal
Over-projected tip	Full transfixion, medial or lateral crural overlay, reducing the footplates

PROBLEM	MANOEUVRE
Under-rotated (drooping) tip	Trim the caudal septum, tongue-in-groove, lateral crural steal or overlay, cephalic trim
Over-rotated tip	Caudal extension graft, medial crural overlay
Internal valve collapse	Spreader grafts, flaring sutures, butterfly graft
External valve collapse	Alar batten grafts, lateral crural strut grafts, alar rim grafts
Saddle nose	Dorsal onlay or reconstruction with septal, ear or *rib cartilage*; rebuild the L-strut

Septal cartilage is the first-choice graft, provided a dorsal and caudal *L-strut* of 10 to 15 mm is left to hold the nose up. **Conchal cartilage** from the ear suits alar grafts, and **costal cartilage** is plentiful for revisions and saddle noses but can warp. Fascia, and diced cartilage wrapped in fascia, smooth the dorsum. The guiding principle is to "preserve and conserve, not resect and regret": reshape with sutures and grafts rather than by removing support.

OPERATIVE GUIDES IN THEATRE 2 operations

Rhinoplasty and surgery of the nasal valve, and septoplasty.

Rhinoplasty and Surgery of the Nasal Valve
Rhinoplasty
Septoplasty—Classic and Endoscopic

2.4 Complications and revision

Between 5 and 15% of rhinoplasties are revised, and revision should wait about a year for the tissues to settle. The named deformities each have a mechanism and a remedy.

A **pollybeak** is fullness above the tip, from an under-resected cartilaginous dorsum, supratip dead space and scar, or loss of tip support; it is treated with steroid injection, trimming or restoring support. A **saddle nose** follows over-resection of the dorsum, septal collapse, a septal abscess or haematoma, or granulomatous disease such as granulomatosis with polyangiitis or cocaine use, and needs grafts.

An **inverted-V deformity** appears when the upper lateral cartilages collapse away from the nasal bones and is corrected with spreader grafts. An **open roof** results from omitting osteotomies after hump removal, and a **rocker deformity** from carrying an osteotomy too high into the thick nasal bone. **Alar retraction and notching** follow over-resection of the lateral crus and need alar rim or composite grafts. Tip bossae, a pinched tip and tip ptosis are the other tip problems.

Functionally, the nose may end up more obstructed through valve collapse, synechiae or residual septal deviation. Septal haematoma or perforation, epistaxis, a rare CSF leak from a high septal or osteotomy fracture, and dissatisfaction complete the list.

3 CHAPTER THREE The ageing face

Upper, middle and lower thirds.

3.1 What ages

Every layer ages. The **skin** develops a thinner dermis with less collagen and elastin, photodamage and wrinkles. The **soft tissues** descend and the fat pads lose volume. The **bone** resorbs, especially at the orbital rims, piriform aperture and maxilla and in the pre-jowl mandible.

The changes show differently in each third. The upper third gains forehead lines and brow ptosis; the middle third shows eyelid laxity and fat bulges, a tear trough and deeper nasolabial folds; and the lower third develops jowls, marionette lines, neck laxity and platysmal bands.

3.2 Brow and forehead lift

The ideal brow begins over the medial canthus and ends on a line from the alar base through the lateral canthus. In women it arches above the orbital rim, peaking between the lateral limbus and the lateral canthus; in men it lies flatter, at the rim.

A **coronal** lift hides the scar well but raises the hairline and leaves the scalp numb, so a **pretrichial** incision is chosen for a high hairline. The **endoscopic** lift uses small

incisions and subperiosteal release with fixation. **Direct** and **midforehead** lifts suit older men with deep forehead lines and patients with facial palsy.

The danger is the **frontal branch of the facial nerve**, which runs in the temporoparietal fascia along *Pitanguy's line*, from 0.5 cm below the tragus to 1.5 cm above the lateral brow. Lateral dissection is therefore kept *deep*, on the deep temporal fascia.

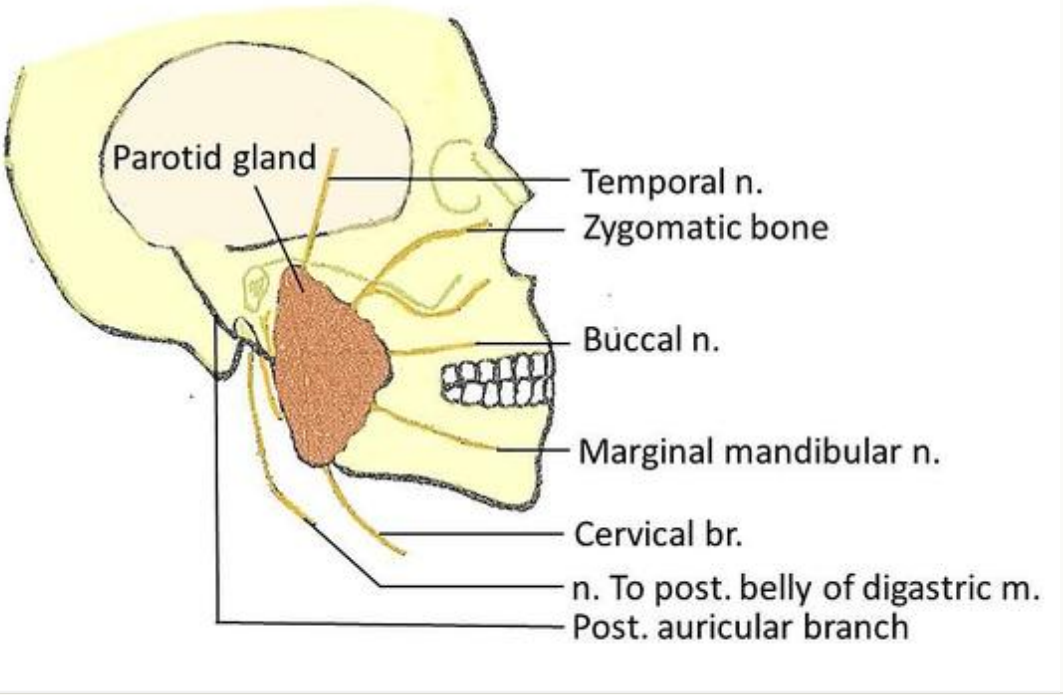


FIGURE 3 Extracranial branches of the facial nerve.
Anatomy atlas · Doss p. 149

3.3 Blepharoplasty

Upper lid blepharoplasty removes excess skin (*dermatochalasis*) with a strip of orbicularis and a conservative amount of fat from the medial and central compartments, through a crease incision about 8 to 10 mm above the lashes. At least *20 mm* of skin must remain between the lash line and the brow, and a ptotic brow, rather than the lid, may be the real problem.

Before surgery, record vision and fields, test for dry eye with the *Schirmer test*, and look for thyroid eye disease, previous surgery and true *blepharoptosis*, which needs levator surgery rather than skin removal.

In the **lower lid**, fat alone is removed through a *transconjunctival* approach, and excess skin through a *subciliary* skin or skin–muscle flap. Lid laxity is tested with the *snap test* and the distraction test, in which more than 6 to 10 mm is abnormal; a lax lid needs a canthopexy or canthoplasty to avoid ectropion. A *negative vector*, with the globe projecting ahead of the cheek, is a warning against removing fat.

The feared complication is a **retrobulbar haematoma**, which can blind; it presents with pain, proptosis and falling vision, and is treated by emergency *lateral canthotomy and cantholysis*, opening the wound, and mannitol, acetazolamide and steroids. The others are lagophthalmos and dry eye, ectropion and scleral show, ptosis, diplopia from inferior oblique injury, and milia.

PAIN AND PROPTOSIS AFTER EYELID SURGERY

Severe pain, a tense proptosed eye and falling vision after blepharoplasty mean a retrobulbar haematoma. Release the sutures and perform a lateral canthotomy and cantholysis at the bedside within minutes, then call ophthalmology. Vision can be lost within about 90 minutes.

3.4 Rhytidectomy (facelift)

The facelift depends on the **SMAS**, the superficial musculoaponeurotic system, continuous with the platysma below and the temporoparietal fascia above. The facial nerve branches run *deep* to it, which is why dissection above the SMAS is safe and dissection beneath it demands care.

Techniques are graded by depth: a **skin-only** lift; **SMAS plication**; **SMAS imbrication** or lateral SMASectomy; the **deep-plane** lift, which goes beneath the SMAS and carries the malar fat to give the strongest midface effect at more risk to the nerves; and the **composite** lift. The neck is treated with liposuction, a midline platysmaplasty and, in selected patients, contouring of the submandibular gland or digastric. Dedo's classification guides the neck: class I minimal deformity, II skin laxity, III fat, IV platysmal banding, V micrognathia and VI a low hyoid.

Haematoma is the commonest complication, at about 3 to 15%, more often in men and hypertensive patients, and must be evacuated promptly. Skin necrosis occurs behind the ear, especially in smokers. The **great auricular nerve** is the nerve most often injured, leaving the ear lobe numb; facial nerve injury, usually of the temporal

or marginal mandibular branch, is usually temporary. Alopecia, a shifted hairline, the pulled-down "pixie ear" and poor scars are the cosmetic failures.

OPERATIVE GUIDES IN THEATRE 3 operations

Brow lift, facelift and blepharoplasty.

Brow Lift Benjamin W. Light, Paul Leong
Brow lift
Face lift
Face lift
Blepharoplasty
Blepharoplasty

4 CHAPTER FOUR The ear

Otoplasty.

4.1 The prominent ear

The prominent ear is the commonest auricular deformity. It results from a deficient *antihelical fold*, a deep *conchal bowl*, or both, sometimes with a protruding lobule. Normally the auriculocephalic angle is about 20 to 30°, the helix sits 15 to 20 mm from the scalp at the top and about 20 mm at mid-ear, and the ear is about 6 cm long and 55% as wide as it is long.

Surgery is usually done at five to six years, when the ear has reached 85 to 90% of adult size and before teasing at school begins. In newborns, *ear moulding* splints in the first weeks correct many deformities without surgery.

4.2 Techniques and complications

Mustardé horizontal mattress sutures create the antihelical fold, and **Furnas** conchomastoid sutures set back the concha; they are placed posteriorly so as not to narrow the canal. Stiff cartilage is weakened by scoring or excision (the Chongchet, Stenström and Converse techniques), and the conchal bowl or lobule may be reduced directly.

Haematoma is the urgent complication, because it leads to perichondritis and a cauliflower ear. Infection and chondritis, suture extrusion and asymmetry also occur. Under-correction or recurrence is the commonest problem overall, and over-correction of the middle third with prominent poles produces the "*telephone ear*".

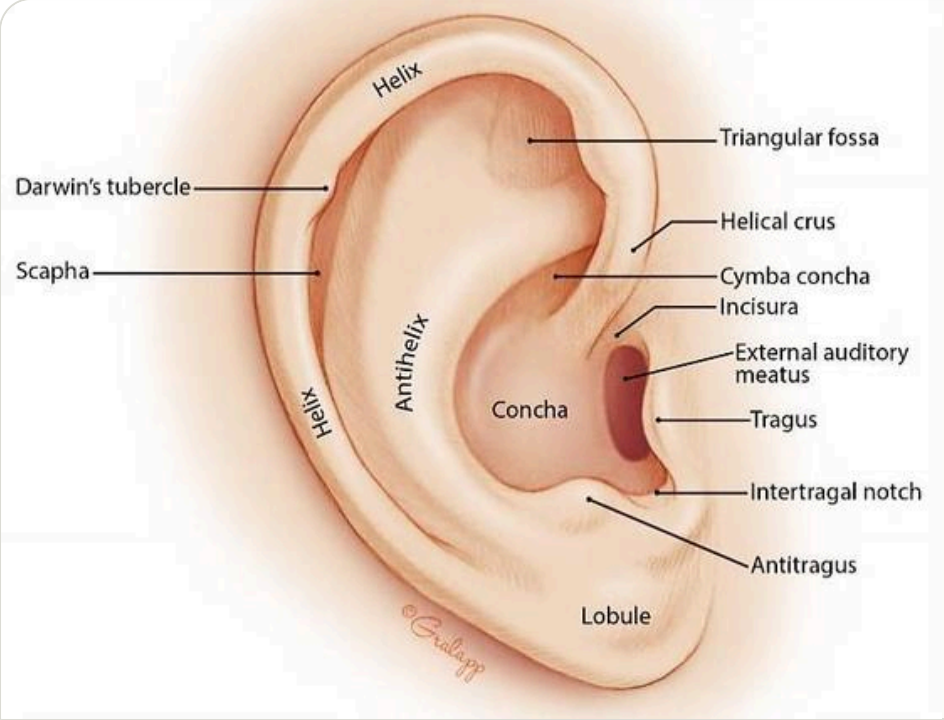


FIGURE 4 Surface landmarks of the auricle.
Anatomy atlas · ENT Steps p. 29

OPERATIVE GUIDES IN THEATRE 1 operation

Otoplasty.

Otoplasty
Otoplasty

5 CHAPTER FIVE Skin and injectables

Resurfacing, toxin, fillers.

5.1 Resurfacing

Every resurfacing method works the same way: it removes the epidermis and injures the papillary dermis, so that new collagen and epidermis form. Injury that reaches

the reticular dermis scars.

Chemical peels are superficial (glycolic acid, Jessner solution, 10 to 25% TCA), medium (Jessner solution followed by 35% TCA) or deep (Baker–Gordon phenol). Phenol is cardiotoxic, causing *arrhythmias*, as well as nephrotoxic and hepatotoxic, so the ECG is monitored, the patient hydrated and one facial unit treated at a time. **Dermabrasion** works mechanically down to the papillary dermis, signalled by pinpoint bleeding, and suits acne scars, rhinophyma and scar revision. Among **lasers**, the ablative CO2 laser penetrates deeper and tightens, while Er:YAG is shallower and heals faster; fractional lasers make microscopic columns of injury that heal rapidly, and non-ablative lasers treat beneath an intact epidermis.

Resurfacing is avoided within 6 to 12 months of isotretinoin (some say two years), and in patients with keloids, previous irradiation or scleroderma. Active herpes needs antiviral prophylaxis, and darker skin types risk pigment change.

5.2 Botulinum toxin

Botulinum toxin blocks the presynaptic release of acetylcholine by cleaving the SNARE proteins, SNAP-25 in the case of type A. It acts within 2 to 3 days, peaks at 1 to 2 weeks and lasts 3 to 4 months.

Cosmetically it treats glabellar lines (corrugator and procerus), forehead lines (frontalis), crow's feet (orbicularis oculi), platysmal bands and the gummy smile. In ENT it treats spasmodic dysphonia, hemifacial spasm and blepharospasm, synkinesis after facial palsy and the opposite side for symmetry, Frey syndrome, sialorrhoea and sialoceles, cricopharyngeal dysfunction, palatal myoclonus and migraine.

Its complications are bruising and headache, *eyelid ptosis* from spread to the levator (treated with apraclonidine drops), brow ptosis, an asymmetric smile and dysphagia. It is avoided in myasthenia gravis, Lambert–Eaton syndrome and other neuromuscular diseases, in patients taking aminoglycosides, and in pregnancy and breastfeeding.

5.3 Fillers and fat

Hyaluronic acid is the commonest filler. It lasts 6 to 18 months and, importantly, can be dissolved with *hyaluronidase*. Calcium hydroxylapatite and poly-L-lactic acid, which stimulates collagen over months, last longer, and permanent fillers such as PMMA and silicone carry a risk of late granulomas. **Autologous fat** is harvested, centrifuged and injected in small aliquots in several layers, with unpredictable survival.

The danger zones are the glabella, nasal dorsum, nasolabial fold and temple, where filler can enter an artery.

VASCULAR OCCLUSION FROM FILLER

- **Skin occlusion:** blanching, pain, then a livedo or dusky pattern after injection. Stop injecting, flood the area with high-dose hyaluronidase (for hyaluronic acid fillers), massage, and apply warm compresses.
- **Visual loss:** retrograde embolism into the ophthalmic artery, from glabellar, nasal or nasolabial injection. This is an ophthalmic emergency.

OPERATIVE GUIDES IN THEATRE 2 operations

Facial resurfacing, and cosmetic botulinum toxin and fillers.

Facial Resurfacing: Peels and Lasers
Cosmetic Botox and Fillers
Botox and fillers

5.4 Scar revision

Scars hide best in *relaxed skin tension lines*, facial creases and the borders of aesthetic units, so revision aims to move them there.

The **Z-plasty** lengthens a scar and changes its direction. With limbs at 30°, 45° and 60° it lengthens the scar by about 25%, 50% and 75%, and turns it by up to 90°, which makes it the tool for contracted scars and for scars lying 30 to 60° off the tension lines. Long scars that run against the lines are broken up with a **W-plasty** or a geometric broken-line closure. Dermabrasion or laser at 6 to 8 weeks refines the result, and steroid injection or silicone controls hypertrophic scars and keloids.

5.5 Hair restoration

Male pattern baldness is androgenetic alopecia, graded by the *Norwood classification*. Minoxidil, finasteride and platelet-rich plasma slow it. Transplantation moves follicular unit grafts from the androgen-resistant occipital scalp, harvested as a strip or by *follicular unit extraction*, with single-hair units placed at the hairline. Transplanted hairs shed after a few weeks and regrow from about three months.

PRESERVATION RHINOPLASTY AND POWERED TOOLS (OXFORD TEXTBOOK)

Removing a dorsal hump in one piece (the Joseph technique) destroys the junction between the nasal bones, upper lateral cartilages and septum, which can lead to middle vault collapse, an *inverted V* deformity and internal valve narrowing; hence the routine use of **spreader grafts** or **spreader flaps** (folding the upper lateral cartilages inwards) after hump reduction, and preserving the mucoperichondrium under the dorsum as a soft-tissue spreader. **Preservation rhinoplasty**, revived in recent years, aims to avoid these problems. As described by Daniel, it has three parts: subperichondrial elevation of the soft tissue envelope, preservation of the nasal ligaments, and **dorsal preservation**, in which the hump is lowered by removing a strip of septum and bone beneath it ("push-down" or "let-down") rather than by resection; it suits a well-shaped dorsum with a hump. **Piezoelectric** instruments and powered burs allow precise osteotomies under direct vision after wide subperiosteal exposure. Every rhinoplasty assessment should include a patient-reported outcome measure and a screening questionnaire for body dysmorphic disorder, at least six standard photographs, and a functional examination of the valves.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Facial proportions** (thirds and fifths) and the main angles.
- **Major and minor tip support mechanisms;** the tripod concept.

- **The internal and external nasal valves:** the Cottle test, spreader and batten grafts.
- **Open versus closed rhinoplasty;** osteotomies; the L-strut.
- **Complications:** pollybeak, saddle nose, inverted V.
- **The frontal branch** and its safe plane; the SMAS.
- **Facelift haematoma;** great auricular nerve injury.
- **Blepharoplasty:** tests, retrobulbar haematoma.
- **Otoplasty:** Mustardé and Furnas sutures; the telephone ear.
- **Botulinum toxin:** mechanism, uses, complications.
- **Filler vascular occlusion and hyaluronidase.**
- **Z-plasty geometry.**

6.2 Common pitfalls

COMMON PITFALLS

- Removing a dorsal hump without osteotomies or spreader grafts (open roof, inverted V).
- An aggressive cephalic trim leading to alar collapse and notching.
- Blepharoplasty without recognising true ptosis, dry eye or a lax lower lid.
- Injecting filler in danger zones without aspiration, slow low-pressure technique, or hyaluronidase available.
- Operating on a patient with body dysmorphic disorder.

Written for this app from Oxford Textbook of Otolaryngology ch. 35, 40, 41, Oxford Specialist Handbook ch. 22, K. J. Lee ch. 54. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

ORBIT & SINUS RELATIONS · TRAUMA

Blow-out Fracture

IMAGING FINDINGS

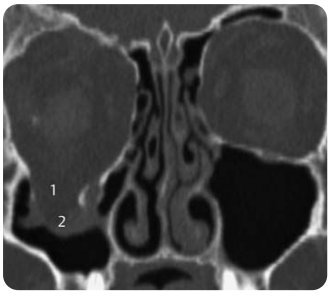
- Fracture of the orbital floor or medial wall with the rim intact
- Herniated fat and muscle; rounded inferior rectus
- Blood in the maxillary sinus; orbital emphysema

TOP DIFFERENTIALS

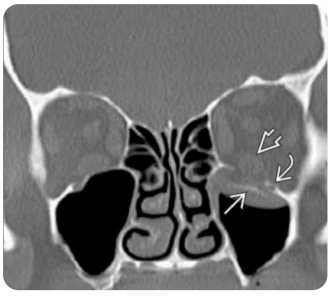
- Dehiscent lamina papyracea
- Old fracture

CHECKLIST

- Muscle entrapment, especially the trapdoor fracture in children
- Defect size (enophthalmos) and retrobulbar haemorrhage



CT, coronal. Traumatic disruption of the bony floor of the orbit (1), and prolapse of orbital fat into the maxillary sinus (2). The eyeball was at a lower…



Coronal bone CT demonstrates a mildly depressed left orbital floor orbital blowout fracture

Koch et al., Diagnostic Imaging: Head…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 980.